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| **Fiscal Year:** | FY 2025 | **Task Last Updated:** | FY 10/20/2025 |
| **PI Name:** | Vanapalli, Siva | | |
| **Project Title:** | Impact of the Gut Microbiome on the Integrative Physiology of Genetically Diverse Invertebrates | | |
| **Division Name:** | Space Biology | | |
| **Program/Discipline:** | | | |
| **Program/Discipline--Element/Subdiscipline:** | | | |
| **Joint Agency Name:** | | **TechPort:** | No |
| **Human Research Program Elements:** | None | | |
| **Human Research Program Risks:** | None | | |
| **Space Biology Element:** | (1) Microbiology<br>(2) Animal Biology: Invertebrate | | |
| **Space Biology Cross-Element Discipline:** | None | | |
| **Space Biology Special Category:** | None | | |
| **PI Email:** | siva.vanapalli@ttu.edu | **Fax:** | FY |
| **PI Organization Type:** | UNIVERSITY | **Phone:** | 806-742-1757 |
| **Organization Name:** | Texas Tech University | | |
| **PI Address 1:** | Chemical Engineering | | |
| **PI Address 2:** | 6th Street and Canton, Mail stop 3121 | | |
| **PI Web Page:** | | | |
| **City:** | Lubbock | **State:** | TX |
| **Zip Code:** | 79409-0001 | **Congressional District:** | 19 |
| **Comments:** | | | |
| **Project Type:** | Flight,Ground | **Solicitation / Funding Source:** | 2020 Space Biology NNH20ZDA001N-SB E.12. Flight/Ground Research |
| **Start Date:** | 03/01/2022 | **End Date:** | 02/28/2026 |
| **No. of Post Docs:** | 1 | **No. of PhD Degrees:** | |
| **No. of PhD Candidates:** | 2 | **No. of Master' Degrees:** | |
| **No. of Master's Candidates:** | 1 | **No. of Bachelor's Degrees:** | |
| **No. of Bachelor's Candidates:** | 1 | **Monitoring Center:** | NASA ARC |
| **Contact Monitor:** | Jones, Harry | **Contact Phone:** | 650.604.5518 |
| **Contact Email:** | harry.jones@nasa.gov | | |
| **Flight Program:** | | | |
| **Flight Assignment:** | NOTE: End date changed to 02/28/2026 per B. Stallsmith/ARC (Ed., 5/20/25)<br>NOTE: End date changed to 02/29/2025 per F. Hernandez/ARC (Ed., 4/25/22) | | |
| **Key Personnel Changes/Previous PI:** | | | |
| **COI Name (Institution):** | Driscoll, Monica Ph.D. ( Rutgers University, New Brunswick )<br>Samuel, Buck S. Ph.D. ( Baylor College of Medicine, Inc )<br>Szewczyk, Nathaniel Ph.D. ( Ohio University ) | | |
| **Grant/Contract No.:** | 80NSSC22K0250 | | |
| **Performance Goal No.:** | | | |
| **Performance Goal Text:** | | | |

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| **Task Description:** | The gut microbiome is a complex community of microbes that exert considerable influence over host physiology, development, metabolism, and predisposition to the disease on Earth. Recent spaceflight investigations in both humans and rodents show that the microbiome community structure is altered under microgravity, posing risk factors for crew health. Thus, understanding the impact of the gut microbiome on long-term physiological changes will be crucial for long-duration space missions to Mars and beyond.<br>The current project is centered around addressing two major knowledge gaps in space biology research. First, fundamental understanding is lacking as to how differences in microbiomes contribute to changes in gut colonization, organ-level physiology, and whole-organism function under microgravity. Second, even though it is recognized on Earth that individual genetic variation can have a large impact when organisms within a species are exposed to new environmental conditions, very little knowledge exists on how genetic diversity within individual species impacts the integrative physiology of organisms when exposed to microgravity since most flight studies to date have focused on genetically homogenous rodent models or cell cultures.<br><br>To address these knowledge gaps, we plan to use C. elegans, which is an established and low-cost invertebrate model for space biology, microbiome studies, and genetic diversity research. We plan to use recently established gut microbial communities to investigate the impact of the gut microbiome on host physiology. In a parallel advance, using genetically diverse wild isolates of C. elegans, we will study host-microbome interactions in spaceflight.<br><br>Our project plan involves testing the following hypotheses in International Space Station (ISS) flight studies with ground study components: (i) Microbial membership of C. elegans gut influences host transcriptional response, tissue-level physiology, and whole-organism function, (ii) Genetic diversity of host influences gut membership of individual microbes, tissue-level physiology, and whole-organism function, and (iii) Insulin signaling pathway play a central role in driving microbiome-induced host response in spaceflight.<br><br>The proposed studies are aligned with the stated strategic goals of NASA Space Biology, which defines over-arching guiding questions focused on integrated biological approaches to understand physiological and molecular mechanisms in living systems that respond to space exploration environments. Pre-biotic and probiotic therapies could be potentially realized from our investigations to improve crew health, along with the dissemination of new flight-tested protocols and molecular characterization tools for the spaceflight community. |
| **Rationale for HRP Directed Research:** | |
| **Research Impact/Earth Benefits:** | The research in this project will quantitate changes in the microbiome as well as in the host in response to space-flight thereby providing detailed understanding of the dynamics of host-microbiome interactions. Additionally, new microfluidic technologies will be developed that facilitate gut-microbiome studies. This new knowledge and capabilities provided by our project can provide significant Earth benefit by contributing to the development of “living medicines” (e.g. probiotics) for treating a variety of human health disorders associated with metabolism, gut, muscle and the nervous system. |
| **Task Progress:** | This "C. elegans Biological Investigation on Microbiome Effect in Space" (CBIOMES) project is a multi-investigator project, bringing teams from four different universities including Texas Tech University, Rutgers University, Baylor College of Medicine, and Ohio University. During the reporting period of 03/01/24 – 02/28/25, there have been significant research accomplishments, as highlighted below.<br>• Optimization of temperature and collection timepoints for worm culture and bacterial colonization in polyethylene (PE) bags.<br><br>Our efforts have focused on optimizing conditions for analyzing bacterial colonization in C. elegans. We examined worm length and bacterial colonization at temperatures of 15°C, 18°C, and 20°C across various timepoints within PE bags. The optimal timeframe for visualizing bacterial colonization in the C. elegans gut, while minimizing offspring production, was identified to be between 7-9 days in PE bags. In parallel, a three-member microbial community was established to accurately replicate the BIGbiome community. [Ed. Note: The BIGbiome community is a model of diverse bacterial species designed to represent the complexity of the wild microbiome of C. elegans.] The combination of Ochrobactrum pituitosum BH3, Lelliottia amnigena JUb66, and Myroides odoratus BIGb0244 was confirmed to best replicate the BIGbiome gut profile at these temperatures. Additionally, optimal conditions for auxin-based sterility were determined. These advancements collectively enhance the study of bacterial interactions within C. elegans.<br><br>• Evaluation of NemaCapsules for C. elegans culture and assess imaging quality.<br><br>-- Significant improvements were made to the auxin exposure protocol prior to worm loading in assays, successfully preventing progeny production. -- Testing of the BIGbiome, alongside the three-biome community and OP50, was conducted through two trials using wild-type (WT) worms in the NemaCapsule lab hardware. These tests assessed worm survival and locomotion under different bacterial diets. In parallel, efforts have been made to develop low-resolution imaging stations and evaluate their imaging quality.<br><br>• Successfully completed science verification testing (SVT). Comprehensive analyses were completed to verify that bag culture conditions, including freezing, shipping, thawing, and fixing samples at various temperatures, consistently produced the required number and quality of animal samples suitable for planned imaging. Representative animals from six fluorescently labeled test strains were imaged to assess high resolution outcomes, successfully meeting the success criteria necessary to proceed to Environmental Verification Testing (EVT).<br><br>• Successful completion of experiment verification testing (EVT). In experiment verification testing, an end-to-end run with all flight samples in PE bags was successfully completed, meeting all required success criteria to advance to the Systems Readiness Flight Review (SRFR). |
| **Bibliography Type:** | Description: (Last Updated: 10/20/2025) |

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| **Articles in Peer-reviewed Journals** | Cope H, Elsborg J, Demharter S, McDonald JT, Wernecke C, Parthasarathy H, Unadkat H, Chatrathi M, Claudio J, Reinsch S, Avci P, Zwart SR, Smith SM, Heer M, Muratani M, Meydan C, Overbey E, Kim J, Chin CR, Park J, Schisler JC, Mason CE, Szewczyk NJ, Willis CRG, Salam A, Beheshti A. "Transcriptomics analysis reveals molecular alterations underpinning spaceflight dermatology." Commun Med (Lond). 2024 Jun 11;4(1):106. https://doi.org/10.1038/s43856-024-00532-9 ; PMID: 38862781; PMCID: PMC11166967 , Jun-2024 |
| **Articles in Peer-reviewed Journals** | Mathyk BA, Tabetah M, Karim R, Zaksas V, Kim J, Anu RI, Muratani M, Tasoula A, Singh RS, Chen YK, Overbey E, Park J, Cope H, Fazelinia H, Povero D, Borg J, Klotz RV, Yu M, Young SL, Mason CE, Szewczyk N, St Clair RM, Karouia F, Beheshti A. "Spaceflight induces changes in gene expression profiles linked to insulin and estrogen." Commun Biol. 2024 Jun 11;7:692. https://doi.org/10.1038/s42003-023-05213-2 ; PMID: 38862620; PMCID: PMC1116698 , Jun-2024 |
| **Articles in Peer-reviewed Journals** | Camera A, Tabetah M, Castañeda V, Kim J, Galsinh AS, Haro-Vinueza A, Salinas I, Seylani A, Arif S, Das S, Mori MA, Carano A, de Oliveira LC, Muratani M, Barker R, Zaksas V, Goel C, Dimokidis E, Taylor DM, Jeong J, Overbey E, Meydan C, Porterfield DM, Díaz JE, Caicedo A, Schisler JC, Laiakis EC, Mason CE, Kim MS, Karouia F, Szewczyk NJ, Beheshti A. "Aging and putative frailty biomarkers are altered by spaceflight." Sci Rep. 2024 Jun 11;14:13098. https://doi.org/10.1038/s41598-024-57948-5 ; PMID: 38862573; PMCID: PMC11166946 , Jun-2024 |
| **Articles in Peer-reviewed Journals** | Mathyk B, Imudia AN, Quaas AM, Halicigil C, Karouia F, Avci P, Nelson NG, Guzeloglu-Kayisli O, Denbo M, Sanders LM, Scott RT, Basar M, Guevara-Cerdán AP, Strug M, Monseur B, Kayisli UA, Szewczyk N, Mason CE, Young SL, Tasoglu S, Costes SV, Beheshti A. "Understanding how space travel affects the female reproductive system to the Moon and beyond." npj Women's Health. 2024 Jun 11;2:20. Review. https://doi.org/10.1038/s44294-024-00009-z , Jun-2024 |
| **Articles in Peer-reviewed Journals** | Rutter LA, Cope H, MacKay MJ, Herranz R, Das S, Ponomarev SA, Costes SV, Paul AM, Barker R, Taylor DM, Bezdan D, Szewczyk NJ, Muratani M, Mason CE, Giacomello S. "Astronaut omics and the impact of space on the human body at scale." Nat Commun. 2024 Jun 11;15:4952. Review. https://doi.org/10.1038/s41467-024-47237-0 ; PMID: 38862505; PMCID: PMC11166943 , Jun-2024 |
| **Articles in Peer-reviewed Journals** | Seylani A, Galsinh AS, Tasoula A, I AR, Camera A, Calleja-Agius J, Borg J, Goel C, Kim J, Clark KB, Das S, Arif S, Boerrigter M, Coffey C, Szewczyk N, Mason CE, Manoli M, Karouia F, Schwertz H, Beheshti A, Tulodziecki D. "Ethical considerations for the age of non-governmental space exploration." Nat Commun. 2024 Jun 11;15:4774. Review. https://doi.org/10.1038/s41467-023-44357-x ; PMID: 38862473; PMCID: PMC11166968 , Jun-2024 |
| **Articles in Peer-reviewed Journals** | Rutter LA, MacKay MJ, Cope H, Szewczyk NJ, Kim J, Overbey E, Tierney BT, Muratani M, Lamm B, Bezdan D, Paul AM, Schmidt MA, Church GM, Giacomello S, Mason CE. "Protective alleles and precision healthcare in crewed spaceflight." Nat Commun. 2024 Jul 22;15:6158. https://doi.org/10.1038/s41467-024-49423-6 ; PMID: 39039045; PMCID: PMC11263583 , Jul-2024 |
| **Articles in Peer-reviewed Journals** | Dierking K, Franke A, Laudes M, Waschina S, Schulenburg H, Kaleta C. "Metabolic model predictions enable targeted microbiome manipulation through precision prebiotics." Microbiol Spectr. 2024 Feb 6;12(2):e0114423. https://doi.org/10.1128/spectrum.01144-23 ; PMID: 38230938 PMCID: PMC10846184 , Feb-2024 |
| **Articles in Peer-reviewed Journals** | Slade L, Bollen SE, Bass JJ, Phillips BE, Smith K, Wilkinson DJ, Szewczyk NJ, Atherton PJ, Etheridge T. "Bisphosphonates attenuate age-related muscle decline in Caenorhabditis elegans." J Cachexia Sarcopenia Muscle. 2023 Dec;14(6):2613-2622. https://doi.org/10.1002/jcsm.13335 ; PMID: 37722921; PMCID: PMC10751425 , Dec-2023 |
| **Articles in Peer-reviewed Journals** | Scott A, Willis CRG, Muratani M, Higashitani A, Etheridge T, Szewczyk NJ, Deane CS. "Caenorhabditis elegans in microgravity: An omics perspective." iScience. 2023 Jul 21;26(7):107189. Review. http://dx.doi.org/10.1016/j.isci.2023.107189 ; PMID: 37456835; PMCID: PMC10344948 , Jul-2023 |